

LESSON 6.4

CONVERTING MASS AND AREA



LESSON INTRODUCTION

MA.7.AR.3.3 Solve mathematical and real-world problems involving the conversion of units across different measurement systems.

LEARNING TARGETS

- I can solve problems involving conversion of mass and area measurements across different measurement systems.

BENCHMARK COHERENCE

BUILDING ON

MA.6.AR.3.5 Solve mathematical and real-world problems involving ratios, rates, and unit rates, including comparisons, mixtures, ratios of lengths, and conversions within the same measurement system.

WORKING TOWARD

MA.8.AR.3 Extend understanding of proportional relationships to two-variable linear equations.

ESSENTIAL QUESTIONS

- What units are used to measure mass, weight, and area in the U.S. customary and metric systems?
- How can conversion rates be used to solve problems?

LESSON NARRATIVE

Students will extend their study of conversions in this lesson with a focus on mass and area conversions. After a Warm-Up that activates prior knowledge in conversions through critical thinking, students convert between and within U.S. customary and metric conversions with a focus on using the conversion rate as a unit rate.

COMPONENT SUMMARY

6.4.1 WARM-UP (~5 MIN.)

Compare echolocation in orcas and 20/20 vision in humans

6.4.3 COLLABORATION (10 MIN.)

Use a given unit to convert both within and between systems for mass and weight

6.4.5 TRY IT (5 MIN.)

Introduction to area in the U.S. customary and metric systems by trying a conversion

6.4.7 WRAP-UP (~5 MIN.)

Reflect metacognitively about understanding and interest in using conversion rates

6.4.2 GUIDED INSTRUCTION (10 MIN.)

Introduction to mass and weight measurements in U.S. customary and metric systems

6.4.4 GUIDED INSTRUCTION (10-15 MIN.)

Reason about different steps in two calculations

6.4.6 YOUR TURN! (5 MIN.)

Practice rounding, comparing unit rates/conversion rates, and determining reasonableness of answers

RESOURCES

GLOSSARY

Key Terms:

- Weight vs. mass

Additional Terms:

- N/A

MATERIALS

- N/A

ADDITIONAL PREPARATION

- Prepare blank double number lines and tables to support multiple means of representations to save them as you Collect and Display.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may struggle with correctly setting up the conversion factor in problems.
- Students may not realize that the structure of the ratio of the conversion rate needs to allow for eliminating a unit in order to convert.

THINGS TO CONSIDER

- Consider having students make a prediction or approximation prior to completing a conversion to use to check for the accuracy or the reasonableness of their answers.
- Students should be allowed to use a calculator to perform the calculations after setting up the conversion factor on paper.

LITERACY AND LANGUAGE ROUTINES

Collect and Display

Every activity provides the opportunity for teachers to represent students' understandings and ideas to help students see as many ways as possible to represent and solve conversion problems.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.4.1 Discussion and Reflection

Note the collaborative analysis of students' work in 6.4.4 Guided Instruction. Make use of this opportunity to provide students with the chance to engage in discussion that reflects on the mathematical thinking of others. This could include comparing the efficiency or accuracy of the methods used.

DIFFERENTIATE INSTRUCTION

Collaborate on an interdisciplinary lesson or project with a science teacher at your school. Students making cross-curricular connections enhance conceptual understanding through a variety of contexts for added fluency and coherence in the STEM field. For example, classes could combine for a co-teacher experience, or your class and the science class could be a Part 1 and Part 2 of one lesson, building on and contributing to each other, or you could borrow a balance or scale to measure mass in real time.

Science Standard SC.8.P.8.2 - Differentiate between weight and mass, recognizing that weight is the amount of gravitational pull on an object and is distinct from, though proportional to, mass.

6.4.1 WARM-UP: ORCA VS. HUMAN

TEACHER MOVES

Consider having students guess or estimate first before calculating to build fluency in planning and decision-making. Students can also refer to the conversion table in Lesson 3 for help in converting from feet to meters.



6.4.1 Warm-Up: Orca vs. Human

Echolocation is the use of sound waves and echoes to navigate and find food, as seen in bats and whales. An orca can "see" objects via echolocation from 150 meters (m) away.



A human with 20/20 vision can clearly see something from 20 feet (ft) away.

- How many times farther can an orca "see" compared to a human?

$$20 \text{ ft} \times \frac{0.305 \text{ m}}{1 \text{ ft}} = 6.1 \text{ m}$$

$$\frac{150}{6.1} \approx 24.6 \text{ times}$$



6.4.2 Guided Instruction: Conversions of Mass and Weight

In the metric system of measurement, the most common units for measuring mass include milligrams (mg), grams (g), and kilograms (kg). In the customary system, the most common units for measurement for weight include ounces (oz), pounds (lb), and tons (t).



Weight vs. Mass

Weight and mass are often used interchangeably, but they are not quite the same, as you'll learn in science. The mass of an object never changes, but its weight changes depending on gravity. For example, your weight on Earth and your weight on the moon would be different, while your mass would remain the same. Since you're in math class here on Earth, mass and weight will be used interchangeably.

- With your partner, discuss some real-world examples of measuring mass.

Answers will vary.

Examples of measuring mass include: 1) measuring mass of a horse; 2) measuring the mass of a car; 3) measuring the mass of a cat

Several common conversion factors for weight and mass are shown.

1 ounce (oz) = 28.35 grams (g)

1 pound (lb) = 454 grams (g)

1 kilogram (kg) = 2.204 pounds (lb)

Each conversion factor can be expressed as a ratio or unit rate. For example, $\frac{454 \text{ grams}}{1 \text{ pound}}$ is the unit rate of grams to pounds.

- Dr. Corlett, a veterinarian, uses weight in kilograms to determine safe dosages of medicine to prescribe for animals.

One of her patients, Opie, is a golden retriever that weighs 85 pounds (lb).

Complete the calculation below to convert Opie's weight to kilograms (kg). Round to the nearest tenth.

$$\frac{85 \text{ lb}}{1} \times \frac{1 \text{ kg}}{2.204 \text{ lb}} = \boxed{38.6} \text{ kg}$$

- Why is it important to include the units in conversion calculations?

Answers will vary.

To ensure you are using a conversion factor that will eliminate the correct unit.

- The table includes information about several of Dr. Corlett's patients.

Name	Breed	Weight (lb.)
Kilo	French Bulldog	16 <i>x</i>
Minnie	Maltese	4 $\frac{1}{2}$ <i>x</i>
Cesar	Irish Terrier	23
Princess	Pug	14.2 <i>x</i>

A certain medication Dr. Corlett uses cannot be prescribed to dogs that weigh less than 10 kg. For which dogs does Dr. Corlett need to be careful not to prescribe this medication?

$$10 \text{ kg} \times \frac{2.204 \text{ lb.}}{1 \text{ kg}} = 22.04 \text{ lb.}$$

Dr. Corlett should not prescribe that medicine to Kilo, Minnie, or Princess. They weigh less than 10 kg.

6.4.2 GUIDED INSTRUCTION: CONVERSIONS OF MASS AND WEIGHT

TEACHER MOVES

Your weight on the Moon is 16.5% of your weight on Earth. This means that a 100-lb. middle school student would weigh 16.5 lbs. on the Moon, roughly the weight of a three-month-old baby or 374 rolls of sushi on Earth.

QUESTIONING

Consider a collaborative lesson with a science teacher. Visit the Unit Guide for ideas.

ENRICHMENT

Does Opie weigh exactly 38.6 kg? (This question reinforces the meaning of the $\approx$ symbol introduced in a prior lesson and used in 6.4.4 Guided Instruction. Consider referencing.)

DIFFERENTIATE INSTRUCTION

Have students substitute the names of their own pets and complete the table based on their estimated or actual weight.

6.4.3 COLLABORATION: CONVERSIONS WITHIN AND BETWEEN SYSTEMS

TEACHER MOVES

Collect and Display

Provide opportunities for students to analyze the work of others and make connections across conversions using the Collect and Display routine. Prompt students to actively describe their work (collect) while another student is capturing those descriptions algebraically (display).

Use chart paper if resources and time permit.

DIFFERENTIATE INSTRUCTION

Consider assigning groups to specific objects and prompt students to compare their work with another group who had the same object as well as analyzing the work of other objects, using Mark-Up strategy to highlight strong student work, misconceptions, or questions (Gallery Walk style).



6.4.3 Collaboration: Conversions Within and Between Systems

Conversion factors that can be used to convert mass between and within measurement systems are shown.

Converting from Customary to Metric
1 ounce = 28.35 grams

Within the Customary System
1 pound = 16 ounces

Converting from Metric to Customary
1 kilogram = 2.204 pounds

Within the Metric System
1 kilogram = 1000 grams

- The weights and masses of four objects are given in the table. Work with your partner to complete the table by using the conversion factors listed. Round to the nearest thousandth when necessary.

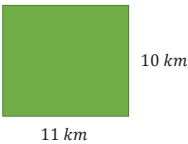
Object	Ounces	Pounds	Grams	Kilograms
Bag of potatoes	80	5	2,268	2.269
Pencil	0.3	0.019	8.505	0.009
iPhone	6.843	0.428	194	0.194
Bicycle	317.376	19.836	9,000	9



6.4.4 Guided Instruction: Conversions of Area

- What does area measure?
The space covered
- What type of units are used to measure area?
Square units

3. Luciana is a wildlife biologist who studies the habitat of the Florida Panther. Florida Panthers are tracked using radio collars to determine the area in which the panther lives. The rectangular habitat of one of the panthers is shown, where the dimensions are given in kilometers (km).



Two students used different methods and conversion factors to find the area of the panther’s habitat in square miles (mi²).

Keyshaun’s Work	Harper’s Work
$1 \text{ mi} = 1.61 \text{ km} *$ $\frac{11 \text{ km}}{1} \times \frac{1 \text{ mi}}{1.61 \text{ km}} \approx 6.832 \text{ mi}$ $\frac{10 \text{ km}}{1} \times \frac{1 \text{ mi}}{1.61 \text{ km}} \approx 6.211 \text{ mi}$ $6.832 \text{ mi} \times 6.211 \text{ mi} \approx 42.43 \text{ mi}^2$	$11 \text{ km} \times 10 \text{ km} = 110 \text{ km}^2$ $1 \text{ mi}^2 \approx 2.5921 \text{ km}^2 *$ $\frac{110 \text{ km}^2}{1} \times \frac{1 \text{ mi}^2}{2.5921 \text{ km}^2} \approx 42.44 \text{ mi}^2$

- Draw an asterisk (*) next to the conversion factor that each student used.
Shown in table.

- Discuss each student’s work with your partner. Then, write a brief description of how each person found the area of the panther’s habitat in square miles.
Answers will vary.

Keyshaun’s Work	Harper’s Work
<i>Keyshawn converted each dimension to miles and then multiplied them to find the area in mi².</i>	<i>Harper found the area in km² and then converted it to mi² using the squared scale factor.</i>

- Keyshaun and Harper used different conversion factors, but still found the same answer. How are their conversion factors related?
 $\left(\frac{1 \text{ mi}}{1.61 \text{ km}}\right)^2 = \frac{1 \text{ mi}^2}{2.5921 \text{ km}^2}$
Harper’s conversion factor is the square of Keyshawn’s conversion factor.
- Why did Keyshaun and Harper use the symbol $\approx$ in their calculations?
Their answers are approximations because they rounded. When converting between systems, the conversion factors are not exact.

6.4.4 GUIDED INSTRUCTION: CONVERSIONS OF AREA

TEACHER MOVES

Students are not being introduced to a new approach or method in conversions. Avoid explaining conversions of area as “different” or “another way” of converting. Instead, have students notice similarities and differences in conversions.

TEACHER MOVES

MA.K12.MTR.4.1: Discussion and Reflection

Note the opportunity for analyzing other students’ work in questions 3a-3d. Provide space for students to analyze independently (a), discuss (b), write (b), and synthesize (c).

DIFFERENTIATE INSTRUCTION

Monitor students to ensure they engage in the discussion before writing and not simultaneously. This will provide multiple entry points for both independent and verbal processors.

6.4.5 TRY IT: CONVERSIONS OF AREA

ENRICHMENT

The term “acre” or concept of an acre might be a barrier of entry for students who are unfamiliar with the term. Consider activating prior knowledge with familiar units to orient students to the content.

- If you are familiar with an acre, what are acres typically used to measure?
- How many square miles are in 1,000 acres? Approximately 1.563
- How many square miles are in 2,000 acres? Approximately 3.125
- How many square miles are in 3,000 acres? Approximately 4.688
- What is a conversion factor you could use? (1.5625 square miles)/(1000 acres)
- How would you use it? Multiply the number of acres by the conversion factor.



6.4.5 Try It: Conversions of Area

Acres are another common unit used to measure large areas of land. An acre is almost the size of a football field, or the size of about 16 tennis courts combined.

1. The Nature Conservancy's Disney Wilderness Preserve covers approximately 11,500 acres of land. Convert the area of the preserve to square miles and to square kilometers. Round to the nearest hundredth when necessary.

$$1000 \text{ acres} = 1.5625 \text{ square miles}$$

$$100 \text{ acres} = 0.405 \text{ square kilometers}$$

Acres	Square Miles	Square Kilometers
11,500	17.97	46.58

$$11,500 \text{ acres} \times \frac{1.5625 \text{ mi}^2}{1000 \text{ acres}} =$$

$$11,500 \text{ acres} \times \frac{0.405 \text{ km}^2}{100 \text{ acres}} =$$



6.4.6 Your Turn!

1. Makayla is a real estate agent. A client is interested in an apartment with a primary bedroom that is rectangular in shape with dimensions of 15 feet (ft) by 11 ft. The client lives in Brazil and has asked for the area of the primary bedroom to be given in square meters (m²). Makayla found the following conversions:

$$1 \text{ ft} = 0.305 \text{ meters (m)}$$

$$10.7584 \text{ square feet (ft}^2\text{)} = 1 \text{ square meter (m}^2\text{)}$$

What is the area of the primary bedroom in square meters? Round to the nearest thousandth.

$$15 \text{ ft} \times 11 \text{ ft} = 165 \text{ ft}^2$$

$$165 \text{ ft}^2 \times \frac{1 \text{ m}^2}{10.7584 \text{ ft}^2} \approx \boxed{15.337 \text{ m}^2}$$

2. Leah lives in Wales. In Wales, body weight is measured using an old customary unit called stone. The stone has been a measurement of mass since before the Romans. Leah is packing to go on vacation and is allowed 25 kilograms (kg) of luggage on the flight. Leah uses her bathroom scale to measure the weight of her luggage. The luggage weighs 3 stones and 3 pounds. Determine if Leah's luggage is over the weight limit.

$$1 \text{ stone} = 14 \text{ pounds}$$

$$1 \text{ kilogram} = 2.204 \text{ pounds}$$

$$3 \text{ stones} \times \frac{14 \text{ lb}}{1 \text{ stone}} = 42 \text{ lb}$$

$$+ 3 \text{ lb}$$

$$45 \text{ lb} \times \frac{1 \text{ kg}}{2.204 \text{ lb}} = 20.417 \text{ kg}$$

Leah's luggage is not over the weight limit.

6.4.6 YOUR TURN!

DIFFERENTIATE INSTRUCTION

Consider prompting students to use a variety of tools and methods to solve problems.

- Encourage students to depict their interpretation of each problem using sketches or drawings and labeling parts of the drawings.
- Provide prompting questions for students to estimate as an entry point. (If the dollar amounts are about the same, would you rather have one pound or five kilograms of something?)

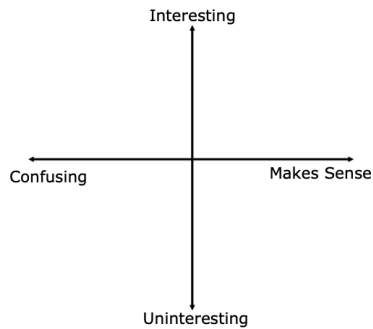


6.4.7 Wrap-Up: Reflection

1. Plot a point on the grid below to indicate how interesting you find the lesson, along with how much sense the lesson made to you, for today's learning target.

I can solve problems involving conversion of mass and area measurements across different measurement systems.

Answers will vary.



6.4.7 WRAP-UP: REFLECTION

DIFFERENTIATE INSTRUCTION

Consider an additional layer of information by prompting students to include the coordinate pair of their self-reflection and include why they chose those values.

Example: (2,5)

I chose 5 because I thought it was cool to think about conversions with 20/20 vision and echolocation to compare. I am only a 2 on solving because I sometimes am still unsure whether to put a value in the numerator or denominator.